

PAVAN J
SENIOR DATA ENGINEER

Email: pavankumarj1199@gmail.com

Phone: +1 (475) 209-2579

LinkedIn: [linkedin.com/in/pavan-j-97abb3219](https://www.linkedin.com/in/pavan-j-97abb3219)

PROFESSIONAL SUMMARY:

- **Senior Data Engineer** with 10+ years of experience in designing, developing, and optimizing large-scale data pipelines, **ETL** workflows, and real-time data processing solutions across cloud and on-premises environments.
- Expertise in **Azure (Azure Data Factory, Azure Synapse, Azure Databricks, Azure SQL Database, Azure Cosmos DB)** and **AWS (AWS Glue, AWS Lambda, Amazon S3, Amazon Redshift, AWS EMR, AWS Step Functions)** for cloud-based data engineering.
- Skilled in building **Azure-based** data pipelines using **Azure Data Factory, Azure Event Hubs, and Azure Service Bus** for real-time and batch data integration.
- Strong experience in monitoring and optimizing cloud-based data workflows using **AWS CloudWatch, AWS X-Ray, and Azure Monitor**, ensuring high system availability and performance.
- Extensive experience in developing and managing **ETL** workflows using **Informatica PowerCenter, Informatica Cloud (IICS), and Informatica Data Quality (IDQ)** for structured and unstructured data processing.
- Proficient in **Python, PySpark, SQL, and Scala**, leveraging libraries such as **Pandas, NumPy, and Spark MLlib** to process and analyze large datasets efficiently.
- Proficient in optimizing **Spark SQL** queries and **Data Frame** operations, enhancing data transformation performance and reducing processing time.
- Hands-on experience in designing and implementing real-time data streaming solutions using **Apache Kafka, Spark Streaming, and AWS Kinesis**, enabling low-latency event-driven architectures.
- Strong expertise in big data technologies, including **Hadoop, HDFS, MapReduce, Hive, Impala, HBase, and Apache Flume**, ensuring scalable and high-performance data processing.
- Proficient in **NoSQL** databases such as **MongoDB, Cassandra, and DynamoDB**, implementing optimized data models and aggregation frameworks for high-throughput applications.
- Experienced in developing **Airflow DAGs** for orchestrating complex **ETL** workflows, ensuring data reliability through robust error handling, logging, and monitoring.
- Experience in **Snowflake** for cloud-based data warehousing, implementing performance optimizations such as clustering, partitioning, materialized views, and **Snow pipe** for seamless data ingestion.
- Expertise in integrating **Power BI, Tableau, and AWS Quick Sight** with big data platforms to develop real-time business intelligence dashboards and analytical reports.
- Expertise in **JIRA, Confluence, and Bitbucket**, streamlining project tracking, collaboration, and version control for large-scale data engineering initiatives.
- Expertise in **CI/CD** automation with **GitHub Actions and GitLab CI/CD**, streamlining data pipeline deployments and minimizing release cycle times.

TECHNICAL SKILLS:

Programming Languages	Python, PySpark, SQL, Scala, T-SQL,
Cloud Platforms	Azure, AWS
Big Data & Streaming	Apache Spark, Hadoop, HDFS, MapReduce, Hive, Impala, HBase, Apache Flume, Apache Kafka, Spark Streaming, AWS Kinesis
ETL & Data Integration	Informatica PowerCenter, Informatica Cloud (IICS), Informatica Data Quality (IDQ), Talend, Apache NiFi, Apache Flume, Azure Data Factory (ADF), AWS Glue
Data Warehousing	Snowflake, Amazon Redshift, Azure Synapse Analytics,
Databases	PostgreSQL, MySQL, MS SQL Server, Oracle, MongoDB, Cassandra, DynamoDB
BI & Visualization	Power BI, Tableau, AWS QuickSight
Workflow Orchestration	Apache Airflow, AWS Step Functions
Machine Learning & Analytics	Spark MLlib, SAS, Python (Pandas, NumPy), Statistical Modeling
DevOps & CI/CD	Git, GitHub, GitLab, Bitbucket, Jenkins, GitHub Actions, GitLab CI/CD
Monitoring & Logging	AWS CloudWatch, AWS X-Ray, Azure Monitor, AWS Config

Project Management & Collaboration	JIRA, Confluence, Microsoft Visio, Lucidchart
Security & Access Control	Role-Based Access Control (RBAC), Data Encryption, OAuth2

PROFESSIONAL EXPERIENCE:

Client: Wells Fargo, San Francisco, California

Aug 2022 – Present

Role: Senior Data Engineer

Responsibilities:

- Developed and managed **Azure SQL Database**, **Azure Cosmos DB**, and **Azure Synapse Analytics** to ensure high availability, security, and performance for structured and unstructured data.
- Designed, developed, and maintained scalable data pipelines and **ETL** processes using **Azure Data Factory**, **Azure Synapse Analytics**, and **Azure Databricks** to enable seamless data integration across enterprise systems.
- Designed, developed, and optimized **ETL** workflows using **Informatica PowerCenter** to extract, transform, and load large datasets from multiple sources.
- Configured and managed **Informatica Cloud (IICS)** and **Informatica Data Quality (IDQ)** to enhance data profiling, cleansing, and validation processes.
- Developed and maintained **ETL** pipelines using **Python** to ingest, transform, and load structured and unstructured data from multiple sources into data warehouses.
- Leveraged **Python libraries** such as **Pandas**, **NumPy**, and **PySpark** to process large datasets efficiently, reducing data processing time and improving overall system performance.
- Utilized **Spark MLlib** for machine learning applications, enabling scalable predictive analytics and clustering.
- Implemented real-time data streaming applications using Spark Streaming and Kafka, processing high-velocity data with low latency.
- Designed and optimized **Spark SQL** queries and **Data Frame** operations, ensuring efficient data transformations.
- Developed **PySpark jobs** to transform and enrich raw data stored in **HDFS**, ensuring high-performance data processing.
- Created **PySpark-based ETL** workflows to extract, transform, and load structured and semi-structured data into data lakes and warehouses.
- Automated data ingestion workflows with **Hadoop Apache Nifi** and **Flume**, enabling seamless real-time and batch data integration.
- Managed and optimized **Hadoop** clusters using **Cloudera** and **Hortonworks** distributions, ensuring high availability and security compliance.
- Designed and implemented **NoSQL** database solutions using **MongoDB**, **Cassandra**, and **DynamoDB** to handle high-velocity and high-volume data.
- Optimized **NoSQL** queries and aggregation frameworks to improve data retrieval efficiency for large-scale applications.
- Implemented error handling, logging, and monitoring mechanisms in **Airflow DAGs** to ensure high reliability and operational efficiency.
- Designed and implemented scalable **ETL** workflows using **Apache Airflow**, orchestrating complex data pipelines with dependencies and retries.
- Designed and implemented **Snowflake-based** data warehouses, enabling scalable and efficient data storage, transformation, and analytics.
- Developed **ETL** pipelines using **Snowflake's** native capabilities, including Snowpipe, Streams, and Tasks, to automate data ingestion and processing.
- Created dynamic reports with real-time data updates by integrating **Power BI** with streaming data sources, ensuring stakeholders receive up-to-date business insights.
- Optimized **Power BI** report performance by implementing data reduction techniques such as aggregations, indexing, and incremental refresh, significantly improving load times.
- Automated **CI/CD** pipelines using **GitHub Actions** and **GitLab CI/CD** to streamline code deployment and integration processes.
- Integrated **JIRA** with **Confluence** and **Bitbucket** to enable seamless collaboration and documentation across development teams.

Environment: Azure SQL Database, Cosmos DB, Synapse Analytics, Data Factory, Databricks, Informatica PowerCenter, Informatica Cloud (IICS), Informatica Data Quality (IDQ), Python, Pandas, NumPy, PySpark, Spark MLlib, Spark Streaming, Kafka, Spark SQL, HDFS, Hadoop, Apache Nifi, Flume, Cloudera, MongoDB, Cassandra, DynamoDB, Apache Airflow, Snowflake, Snowpipe, Streams, Power BI, GitHub Actions, GitLab CI/CD, JIRA, Confluence, Bitbucket.

Client: Allstate, Northbrook, IL

Feb 2019 – Jun 2022

Role: Data Engineer

Responsibilities:

- Monitored and optimized **AWS-based** data workflows using **AWS CloudWatch**, **AWS X-Ray**, and **AWS Config**, ensuring system reliability and performance.
- Engineered scalable data pipelines using **AWS Glue**, **AWS Lambda**, and **AWS Step Functions**, ensuring seamless data ingestion, transformation, and processing.
- Developed reusable **Informatica mappings**, sessions, and workflows, ensuring efficient data movement and transformation across heterogeneous systems.
- Designed and developed scalable **ETL** pipelines using **Python** to process large volumes of structured and unstructured data for analytics and reporting.
- Wrote **SQL** queries and **Python scripts** for efficient data extraction from relational databases, improving query performance and reducing load times for business reporting dashboards.
- Tuned **Spark job** performance by optimizing shuffle operations, caching strategies, and parallel execution settings, reducing processing time.
- Developed **Spark-based ETL** pipelines, integrating with data lakes, **NoSQL** databases, and relational databases to enable seamless data ingestion.
- Developed and maintained real-time data streaming solutions using **PySpark** and **Apache Kafka** for event-driven processing.
- Developed **Sqoop jobs** to efficiently import and export data between **Hadoop**, relational databases, and cloud storage.
- Developed and optimized large-scale data pipelines using **Hadoop Distributed File System (HDFS)**, ensuring efficient data storage and retrieval for analytical processing.
- Integrated **Hadoop** ecosystem components such as **Hive**, **Impala**, and **HBase** to enable efficient querying and analysis of big data.
- Implemented role-based access control and encryption mechanisms to secure **NoSQL** data storage and ensure compliance with regulations.
- Developed **Airflow** plugins and custom operators to extend functionality and integrate with external systems.
- Optimized **Snowflake** performance by leveraging clustering, partitioning, and materialized views for faster query execution.
- Developed complex **SQL** queries and stored procedures in Snowflake to support business intelligence.
- Automated **Tableau** data refreshes and scheduled report updates using **Tableau Server** and **Tableau Online**, ensuring users always have access to the latest business data.
- Developed real-time dashboards using **Tableau's** Live Connection feature, enabling business teams to monitor operational metrics and key performance indicators with minimal latency.
- Maintained detailed **Git** commit history and documentation to track changes, enable rollback, and ensure reproducibility of software versions.
- Defined and configured **JIRA** workflows, issue types, and custom fields to optimize project tracking.

Environment: AWS CloudWatch, X-Ray, Config, Glue, Lambda, Step Functions, Informatica, Python, SQL, Spark, PySpark, Apache Kafka, Sqoop, Hadoop Distributed File System (HDFS), Hive, Impala, HBase, NoSQL, Airflow, Snowflake, Tableau, Git, JIRA.

Client: Johnson & Johnson, NJ

May 2016 – Jan 2019

Role: Data Engineer

Responsibilities:

- Configured and monitored **Azure Event Hubs** and **Azure Service Bus** for real-time data streaming and message processing, enabling seamless communication between distributed systems.
- Built scalable **ETL** workflows using **Azure Data Factory (ADF)**, incorporating data cleansing, transformation, and validation for accurate and consistent data ingestion.
- Integrated **Informatica Power Exchange** with mainframes, databases, and cloud environments to enable real-time and batch data processing.
- Optimized **Python-based** data engineering workflows using memory-efficient algorithms, parallel computing, and caching mechanisms.
- Developed and optimized **Spark-based ETL** pipelines to process large-scale structured and unstructured data, improving data transformation performance and reducing execution time.
- Converted **SQL-based** transformations into **PySpark** operations to enhance execution speed and handle large-scale data workloads.

- Designed and implemented **Hadoop-based** data lakes, enabling large-scale data ingestion, processing, and storage for enterprise analytics.
- Designed and managed **Hadoop, MapReduce jobs** to process terabytes of structured and semi-structured data, improving data transformation performance and reliability.
- Migrated relational database workloads to **NoSQL** databases, leveraging appropriate data modeling techniques for performance gains.
- Configured role-based access control (**RBAC**) and authentication mechanisms in **Airflow** to enforce security.
- Integrated **Power BI** with **Microsoft Power Automate** to trigger workflows based on data changes, enabling automated alerts and actions for business-critical scenarios.
- Developed custom **Power BI** visuals and templates tailored to business needs, enabling effective storytelling and advanced analytical capabilities for executive reporting.

Environment: Azure Event Hubs, Service Bus, Data Factory (ADF), Informatica Power Exchange, Python, PySpark, Spark, SQL, Hadoop, MapReduce, NoSQL, Airflow, Role-Based Access Control (RBAC), Power BI, Microsoft Power Automate.

Client: Target, MN

Jun 2013 – Feb 2016

Role: Data Analyst

Responsibilities:

- Developed and optimized **Amazon S3-based** data lakes, enabling cost-efficient storage, retrieval, and analytics for large-scale data processing needs.
- Managed **Amazon Redshift** clusters, implementing performance tuning techniques such as distribution keys and sort keys to enhance query performance.
- Implemented statistical models using **SAS** procedures to analyze large datasets and generate predictive insights for business decisions.
- Processed and analyzed large datasets in **Excel**, integrating external data sources such as **SQL** databases and **APIs** for enhanced insights.
- Deployed **Tableau** dashboards on web portals and embedded analytics platforms, integrating them with enterprise applications to enhance data accessibility and user engagement.
- Tuned **Hadoop** performance by optimizing block size, compression, and resource allocation in **YARN** for better efficiency.
- Developed **API-driven NoSQL** data access layers to support real-time applications, event-driven architectures, and microservices.
- Optimized **Python scripts** for real-time data processing and batch processing using multiprocessing and threading techniques.
- Designed **Tableau Prep** workflows to clean, reshape, and prepare large datasets before visualization, significantly improving report accuracy and consistency.

Environment: Amazon S3, Redshift, SAS, Excel, SQL, APIs, Hadoop, YARN, NoSQL, Python, Tableau Prep.

EDUCATION:

Jawaharlal Nehru Technological University Hyderabad - 2011

Bachelor of Engineering in computer science